

MD ENAMUL HOQ

Ph.D. Candidate, Biomedical Informatics (Imaging) · Translational Imaging AI for Oncology

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RESEARCH INTERESTS

Translational imaging informatics and computer vision for oncology, spanning radiology and digital pathology: large-scale imaging pipelines, self-supervised foundation models, and multimodal learning across CT and whole-slide imaging. My work centres on representation learning, DICOM-aware preprocessing, dataset curation and annotation, CT quality control, explainable AI, and clinically aligned evaluation — linking imaging phenotypes to histomorphological and molecular characterisation to build robust, interpretable and reproducible systems for cancer diagnosis, monitoring, prognosis and clinical translation.

EDUCATION

Ph.D., Biomedical Informatics (Imaging Track)

Expected May 2026

University of Arkansas for Medical Sciences (UAMS), Little Rock, AR

Dissertation: Foundation Models for Lung Cancer Screening CT (LCS-DINO, native-resolution 2D).

M.S., Engineering Physics

December 2020

Southeastern Louisiana University, Hammond, LA

Thesis: Image correlation for LDPE film deformation — computational imaging in MATLAB.

RESEARCH EXPERIENCE & APPOINTMENTS

Graduate Research Assistant (Ph.D. Researcher)

07/2021 – present

University of Arkansas for Medical Sciences (UAMS), Little Rock, AR

- Built an end-to-end lung cancer screening pipeline — DICOM/PACS-compliant curation, embedding generation, training and patient-level evaluation — across 20M+ CT slices.
- Managed large-scale cohorts in TCIA and the NCI Imaging Data Commons: cohort definition, manifest generation, DICOMweb/gsutil transfer, BigQuery access, metadata harmonisation (DICOM tags, RadLex, SNOMED) and research governance (DUAs, PHI checks).
- Designed threshold-aware evaluation (ROC/AUC, PR/F1, prevalence-matched cut-points) reflecting real screening prevalence; reduced false-positive follow-ups in experimental cohorts.
- Scaled multi-GPU training and inference on A100 clusters with Slurm; automated experiment tracking, versioning and report generation for reproducible, auditable results.
- Initiated and lead a longitudinal phantom-based study of the reliability and drift of commercial AI lung-nodule detection software (Riverain VS-CT) at the Winthrop P. Rockefeller Cancer Institute, under an IRB-approved prospective protocol.
- Developed a MATLAB phantom-modelling pipeline that injects synthetic pulmonary nodules into real DICOM CT volumes, with a database and analysis workflow for detection accuracy, measurement reproducibility and longitudinal stability.
- Collaborated with 15+ radiologists and clinicians to translate model outputs into clinically interpretable metrics and structured reporting.
- Mentored three Ph.D. students and one M.D. student in GPU/HPC workflows, DICOM/PACS handling, experimental design and reproducible research.

Site Lead — Federated LDCT AI Collaboration (Site 2)

06/2025 – present

UAMS, Little Rock, AR · with Moffitt, Vanderbilt, UCLA and the University of Pittsburgh

- Lead UAMS as Site 2 in a five-institution federated learning programme for low-dose CT lung cancer screening using NVIDIA FLARE.

- Architected privacy-preserving training workflows — secure model updates and cross-site evaluation protocols — that improve detection performance without sharing raw PHI.
- Curate and harmonise multi-site imaging datasets with leakage-safe splits and AI-ready pipelines for cross-institution generalisability.
- Approach recognised as Champion in three of five categories at the CMU-NVIDIA Federated Learning Hackathon, 2026.

AI & Informatics Research Intern

02/2025 – 07/2025

Mayo Clinic, Rochester, MN

- Developed a multimodal pathology retrieval and auto-captioning framework using vision foundation models and LLMs — top-ten poster, Mayo Clinic AI Summit 2025.
- Raised semantic alignment of generated captions to above 0.82 cosine similarity via a hybrid rule-based and LLM summarisation pipeline, and introduced retrieval-guided generation to reduce unsafe and hallucinated captions.
- Co-led a workshop on fine-tuning and deploying vision-language models in clinical research workflows.

Graduate Research & Teaching Associate; Lab Manager

08/2019 – 12/2020

Southeastern Louisiana University, Hammond, LA

- Developed MATLAB pipelines for digital image correlation (LDPE deformation) with visualisation and modelling tools.
- Supported 13+ researchers with debugging and optimisation; instructed Physics 101/102 laboratory courses.

Research Intern — Multimedia Lab

09/2018 – 02/2019

National Chung Cheng University, Chiayi, Taiwan

- Developed depth-estimation algorithms for 2D→3D video conversion, improving perceived depth and visual quality.

R&D Engineer

08/2017 – 07/2018

TN Solution — Karlsruhe, Germany & Ho Chi Minh City, Vietnam

- Led image/signal-processing and IoT projects; improved PCB fault diagnosis through noise reduction and feature extraction.

PUBLICATIONS & SCHOLARLY WORK

Peer-reviewed publications

1. **Hoq ME**, Tarbox L, Johann D Jr, Larson-Prior L, Prior F. Harnessing native-resolution 2D embeddings for lung cancer classification: a feasibility study with the RAD-DINO self-supervised foundation model. *Journal of Imaging Informatics in Medicine*, 2025. [doi:10.1007/s10278-025-01748-4](https://doi.org/10.1007/s10278-025-01748-4)
2. Alfasly S, Uegami W, **Hoq ME**, Alabtah G, Tizhoosh HR. Semantic and visual crop-guided diffusion models for heterogeneous tissue synthesis in histopathology. *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. openreview.net/forum?id=yNVDkAjGjw
3. **Hoq ME**, Uegami W, Alfasly S, Alabtah G, Malakshan S, Kazemi A, Schmitgen AT, Prior F, Tizhoosh HR. Retrieval-guided generation for safer histopathology image captioning. *Ital-IA 2026, CEUR Workshop Proceedings* (ISSN 1613-0073), 2026. [arXiv:2605.00893](https://arxiv.org/abs/2605.00893)

Accepted / in press

4. **Hoq ME**, Larson-Prior L, Prior F. Virtual-Eyes: quantitative validation of a lung CT quality-control pipeline for foundation-model cancer-risk prediction. *Medical Imaging with Deep Learning (MIDL)*, accepted 2026. [arXiv:2512.24294](https://arxiv.org/abs/2512.24294)

Preprints & technical reports

5. **Hoq ME**, Prior F. Artificial intelligence for low-dose CT lung cancer screening: a systematic review of deep learning, input dimensionality, workflow integration and foundation models. *Preprints.org*, 2026. [doi:10.20944/preprints202606.1208.v1](https://doi.org/10.20944/preprints202606.1208.v1)
6. Mu J, Karna AK, Kumar Koushik T, Winchell J, Yang TJ, Zhang CM, Baker J, Hagen E, **Hoq ME**, Kim K, ... Busby B. Towards federated learning across biobanks: prototype software from the 2026 Carnegie Mellon University-NVIDIA hackathon. *OSF Preprint*, 2026.

Published abstracts & conference proceedings

7. **Hoq ME**, Uegami W, Alfasy S, Malakshan SR, Alabtah G, Kazemi A, et al. From CNN to vision foundation models and LLMs: a multimodal framework for pathology image retrieval and auto-captioning. *Mayo Clinic Proceedings: Digital Health* 3(4), 2025.
8. Kolb S, Sunkara H, Chavane C, **Hoq ME**, Ali M, Xu H, Kashyap R. Renal replacement therapy following IV contrast administration in critically ill patients with CKD. *Critical Care Medicine* 53(1), 2025.
9. Barnes B, **Hoq ME**, Binette E, Lee S, Arumairaj A. Impact of an elevated fluid-accumulation index in septic patients with HFrEF: a MIMIC-IV analysis. *Critical Care Medicine*, 2026.
10. **Hoq ME**, Prior F. Optimizing and harnessing 20 million NLST CT lung-screening images for robust foundation-model training. *NIH Early Detection Research Network (EDRN)*, 2024.
11. **Hoq ME**, Yoshida S, Jung H. Measurement of thickness and preservation of thin films at maximum wrapping speed. *APTEC Annual Meeting*, 2019.

Thesis

12. **Hoq ME**. An investigation of image correlation for real-time monitoring of wrapping-film deformation. M.S. thesis, Southeastern Louisiana University, 2020.

SELECTED INVITED TALKS & POSTERS

- UAMS Radiology Retreat, 2026 — Synthetic CT phantom for nodule-detection validation.
- UAMS Winthrop P. Rockefeller Cancer Institute, 2026 — From CT quality control to structured radiology reporting and validation.
- NCI Early Detection Research Network (Caltech), 2024 — Optimizing and harnessing 20M NLST CT lung-screening images.
- APTEC Annual Meeting, 2019 — Digital image correlation for LDPE deformation (audience 200+).

GRANTS & LARGE-SCALE RESEARCH INITIATIVES (CONTRIBUTOR)

- NCI U24 CA215109 — The Cancer Imaging Archive (TCIA) sustainment and scalability. Contributor to imaging-data infrastructure supporting multi-institutional AI research and data sharing.
- UAMS Team Science Award (\$100K) — lead and main contributor to an interdisciplinary initiative on clinical AI and imaging analytics.
- UAMS Team Science Award (\$40K) — lead and main contributor to a collaborative programme supporting translational imaging and data-driven clinical studies.
- EU EuCanImage (SEP 210631720; €805,479) — contributor to an international imaging-AI ecosystem for multi-centre data integration and cancer research.
- MOTIE Korea International R&D Program (P0006842) — contributor to global technology collaboration in imaging and data-driven innovation.

HONORS & AWARDS

- Champion, three of five categories — CMU-NVIDIA Federated Learning Hackathon, 2026.
- Top-ten poster — Mayo Clinic AI Summit, 2025.
- Second place — Datathon.org, Emory University School of Medicine, 2024.
- Travel grants — SIIM (2025), SCCM (2024–2025), HITLAB (2024).
- Merit Scholarship — World University of Bangladesh.

PROFESSIONAL SERVICE & LEADERSHIP

Editorial & peer review

- Ethics Reviewer — NeurIPS, 2026.
- Program Reviewer — Society for Imaging Informatics in Medicine (SIIM) Annual Conference, 2025–present.
- Ad hoc reviewer — Frontiers in Medicine, 2024–present (6+ manuscripts); Journal of Thoracic Disease.

- Poster judge — UAMS Summer Research Program, Winthrop P. Rockefeller Cancer Institute, 2024 (17 posters).

Conference & project leadership

- Area Chair, Challenges Track — Medical Image Computing and Computer Assisted Intervention (MICCAI), 2026.
- Site Lead (UAMS) — multi-institutional federated learning project for LDCT lung cancer screening (Moffitt, Vanderbilt, UCLA, U. Pittsburgh).
- Workshop co-lead — Practical AI in Healthcare: fine-tuning and deployment with open-source tools, Mayo Clinic, 2025.

TEACHING & MENTORING

- Research mentorship — Medical Imaging AI, UAMS (2023–present). Mentored three Ph.D. students and one M.D. student in imaging AI, GPU/HPC workflows, DICOM/PACS handling, experimental design and reproducible research.
- Graduate Teaching & Research Associate — Southeastern Louisiana University (2019–2020). Instructed Physics 101/102 laboratories and lectures; supported curriculum delivery and training in experimental methods.
- Instructor & mentor — foundation-year physics and mathematics (2010–2017). Taught and mentored 10,000+ pre-university students; developed problem-based learning materials and exam-preparation frameworks.

TECHNICAL SKILLS

Machine learning: Model development, training, validation and deployment; supervised, self-supervised and representation learning; classification, detection and quantitative image analysis; imbalanced-data learning; transfer learning and fine-tuning

Architectures: 2D and 3D CNNs; Vision Transformers; self-supervised foundation models (RAD-DINO, DINOv2); diffusion models; multimodal vision–language models

Imaging & standards: CT; chest radiography foundation models; digital pathology (WSI); DICOM, PACS, DICOMweb; volumetric and density quantification; RadLex, SNOMED; 3D Slicer, ITK-SNAP; TCIA, TCGA, NCI Imaging Data Commons

Programming: Python (primary), C/C++, Java, MATLAB

Frameworks: PyTorch, TensorFlow, Keras; NVIDIA FLARE (federated learning)

Evaluation: ROC/AUC, sensitivity/specificity, precision/recall, PR/F1; calibration; threshold and cut-point optimisation; reproducibility, drift and stability analysis

Engineering practice: Git and collaborative development; code review; testing; experiment tracking and automated reporting; data/model/code versioning and traceability

Compute & cloud: Linux; Slurm; HPC with A100 multi-GPU clusters; Google Cloud Platform; AWS; BigQuery

Data engineering: Large-scale imaging pipelines; leakage-safe patient-level splitting; multi-institutional harmonisation and quality control; metadata normalisation; data governance (DUAs, PHI de-identification)

Regulatory & quality: FDA 510(k) and De Novo-oriented documentation; Good Machine Learning Practice (GMLP); reproducible, auditable validation; clinical-deployment and continuous-monitoring evaluation; IRB-approved prospective validation

SELECTED DATASETS & SOFTWARE

- NLST-derived CT screening cohort (20M+ slices). Internal UAMS-curated dataset: preprocessing, DICOM normalisation, leakage-safe patient-level splits and RAD-DINO embedding pipelines for large-scale training and evaluation.
- Multimodal pathology retrieval and auto-captioning pipeline. Vision foundation-model embeddings, rule-based labelling and LLM summarisation, with interactive tools for retrieval evaluation and semantic alignment.
- HunCRC and ARCH datasets. Generated embeddings and performed leave-one-out retrieval analysis for histopathology image similarity and representation benchmarking.

CERTIFICATIONS & MEMBERSHIPS

- Certified in Lung Cancer Screening Education — American College of Radiology (ACR), 2024.
- National Imaging Informatics Course (NIIC) — SIIM / AAPM / RSNA.

- 20+ professional certifications — NVIDIA, Google, Coursera (AI, deep learning, cloud computing, data science).
- Member — The MICCAI Society; Society for Imaging Informatics in Medicine (SIIM); Radiological Society of North America (RSNA); American College of Radiology (ACR); Society of Critical Care Medicine (SCCM).

REFERENCES

- Prof. Dr. Fred W. Prior — Distinguished Professor & Chair, Department of Biomedical Informatics, UAMS · FWPrior@uams.edu · +1 (314) 303-2485
- Dr. Lawrence Tarbox — Associate Professor & Head of HPC, Department of Biomedical Informatics, UAMS · lrtarbox@uams.edu · +1 (501) 603-1766
- Dr. Donald J. Johann Jr. — Professor (M.D.), Cancer Therapeutics, UAMS · djjohnann@uams.edu · +1 (501) 526-6990